

SIT723 Research Techniques & Applications

Task 3.3C: Select Research Methods

Methodology Plan

Dataset Comparison

Dataset	Size / Type	Annotations	Unique Features	Availability	Research Use Cases
MedMentions ST21pv Primary dataset n = 550	4,392 PubMed abstracts; 350,000+ UMLS-linked mentions. ST21pv subset with 21 preferred semantic types.	UMLS Concept Unique Identifiers (CUIs) for each entity mention. Expert annotation with 97.3% precision.	Formal ontological grounding via UMLS Metathesaurus. Enables concept-level semantic entropy evaluation.	Publicly available under CC0 licence. Preprocessing: instances matched against scispaCy UMLS linker (confidence ≥ 0.50); stratified sampling across three semantic type categories.	• Primary dataset for concept-level semantic entropy evaluation. • Stratified sample: Disease and Syndrome (n~185), Pharmacological Substance (n~185), Procedure and Clinical Finding (n~180). • Supports RQ1, RQ2, and RQ3.
SQuAD 2.0 Secondary dataset n = 200	100,000 answerable + 50,000 unanswerable	Answer spans with start positions.	Includes unanswerable questions requiring	Publicly available under CC BY-SA 4.0.	• General-domain cross-domain validation.

Dataset	Size / Type	Annotations	Unique Features	Availability	Research Use Cases
	e questions. Stratified answerable sample of n = 200 used.	Four answer types: named entity, date/numeric, short descriptive phrase, clausal answer.	abstention. Enables cross-domain validation of semantic stability beyond clinical text.		• Tests whether semantic stability findings generalise beyond clinical NLP. • Supports RQ3 generalisation condition.
BioASQ Task B Phase A Tertiary dataset n = 150 Years: 2021-2023	Biomedical QA challenge linked to PubMed literature. Factoid (~90) and yes/no (~60) question types.	Gold answers with supporting PubMed article identifiers. Expert-validated answer spans.	Intermediate biomedical QA setting between clinical concept normalisation (MedMentions) and general-domain QA (SQuAD 2.0). Enables domain-continuum analysis.	Publicly available at bioasq.org for registered participants.	• Biomedical QA validation. • Domain-continuum analysis: MedMentions (clinical NER) to BioASQ (biomedical QA) to SQuAD 2.0 (general QA). • Supports RQ3.

Required Resources

To address the research questions, the following resources will be used:

Datasets

MedMentions ST21pv: 4,392 PubMed abstracts with 350,000+ UMLS-linked entity mentions. Stratified sample of $n = 550$ instances across three semantic type categories will be used for the primary analysis.

Publicly available under CC0 licence with no data access request required.

SQuAD 2.0: $n = 200$ answerable questions stratified across four answer types (named entity, date/numeric, short descriptive phrase, clausal answer). Publicly available under CC BY-SA 4.0.

BioASQ Task B Phase A (2021-2023): $n = 150$ questions (factoid: ~ 90 ; yes/no: ~ 60). Publicly available at bioasq.org for registered participants.

Computing Infrastructure

Two NVIDIA RTX A4000 GPUs (16 GB VRAM each; 32 GB total). GPU-1 allocated to encoder-only models (BERT-base, BioBERT, PubMedBERT, FLAN-T5-base) GPU-2 allocated to large generative models (FLAN-T5-XXL, BioMistral-7B) with INT8 quantisation via bitsandbytes.

200 GB SSD local storage for model checkpoints, perturbation datasets, and output logs.

Python environment with: PyTorch, HuggingFace Transformers, bitsandbytes, scispaCy (v0.5.4), NLTK WordNet 3.1, Helsinki-NLP/opus-mt back-translation models, scikit-learn, scipy, statsmodels, lme4 (R), simr (R for power analysis).

Estimated wall time: 26-43 hours across all model inference and perturbation generation. Storage requirement: ~ 49 GB of the 200 GB SSD capacity.

Software and Tools

scispaCy entity linker (linker_name='umls', version 0.5.4) for UMLS CUI mapping.

MetaMap (NLM, version 2023) as independent second UMLS linker for cross-linker reliability validation (target: Cohen's kappa ≥ 0.75 with scispaCy).

SapBERT (cambridgeltl/SapBERT-from-PubMedBERT-fulltext) for embedding-based disambiguation in the CUI mapping protocol.

DeBERTa-v3-large-mnli-fever-anli for NLI-based output clustering in QA tasks.

cross-encoder/nli-MiniLM2-L6-H768 for Gate G2 bidirectional entailment validation in the quality pipeline.

GPT-2-medium (345 MB; CPU inference) for grammaticality checking in Gate G4.

OSF (Open Science Framework) for pre-registration of all confirmatory analyses.

Selected Research Methods

The research will use semantic entropy evaluation framework based on UMLS. The key methodological novelty is that instead of the approximate NLI-based output clustering used by L. Kuhn et al. (2023), a formal UMLS CUI equivalence criterion will be used, which allows measuring semantic stability of concepts in clinical NLP. It will integrate four key families of methods: meaning-preserving perturbation generation, automated quality validation, model inference in five LLMs, and semantic entropy computation and subsequent pre-registered statistical testing.

Models Evaluated

All models will be evaluated in standardised zero-shot or structured-prompt settings with no task-specific fine-tuning. Greedy decoding (temperature = 0.0) is used for all generative models to eliminate sampling stochasticity as a confound.

Encoder-only models (GPU-1, CUI-ranking entropy):

BERT-base (110M parameters): General-purpose foundational encoder baseline.

BioBERT (110M parameters): Continual pretraining on PubMed + PMC data; primary domain-specific model in within-scale comparison for RQ3.

PubMedBERT (110M parameters): From-scratch biomedical pretraining on PubMed; additional domain-specific encoder for RQ3.

Generative models (GPU-2, text-generation entropy, INT8 quantisation):

FLAN-T5-base (250M): General-purpose instruction-tuned seq2seq model.

FLAN-T5-XXL (11B, INT8, ~11 GB VRAM): Large-scale general-purpose counterpart to BioMistral-7B for within-scale H3 comparison.

BioMistral-7B (7B, INT8, ~8 GB VRAM): Open-source biomedical LLM; primary domain-specific large model for within-scale H3 comparison.

Meaning-Preserving Perturbation Framework

Each input instance is transformed into $k = 8$ meaning-preserving perturbations using four methods, yielding nine total inputs per instance (one original plus eight perturbations). All model checkpoints are pinned to specific HuggingFace revision hashes documented in the released configuration file.

Back-translation:

Helsinki-NLP/opus-mt-en-de + opus-mt-de-en (EN to DE to EN).

Helsinki-NLP/opus-mt-en-fr + opus-mt-fr-en (EN to FR to EN).

Beam search $k = 5$; batch size 32; max output length 2x source length.

Controlled paraphrase:

tuner007/pegasus_paraphrase -- temperature 1.5, top-p 0.9.

Vamsi/T5_Paraphrase_Paws -- select from 3 candidates by cosine similarity to original.

Synonym substitution:

Medical: UMLS Metathesaurus REST API v2023AA (atom-level synonymy).

General: NLTK WordNet 3.1 with POS-constrained substitution (substituted tokens must share POS tag with originals).

Two variants: 1-word and 2-word substitution.

Syntactic reordering:

SpaCy en_core_web_trf dependency parser + deterministic rule-based transformation grammar.

Two variants: active to passive voice conversion and clause fronting.

Requires transitive verb with explicit subject and object.

Six-Gate Automated Quality Validation Pipeline

All generated perturbations must pass six sequential quality gates before inclusion. Gates are designed to guard against distinct, non-overlapping failure modes. Up to three regeneration attempts are made per rejected perturbation slot. All thresholds are calibrated empirically on UMLS ground-truth pairs before data collection.

G1: Embedding similarity (threshold ≥ 0.85):

Cosine similarity between original and perturbation sentence embeddings.

SapBERT for medical instances; all-MiniLM-L6-v2 for general instances.

Guards against semantic drift.

G2: Bidirectional entailment (threshold ≥ 0.80 in both directions):

Model: cross-encoder/nli-MiniLM2-L6-H768 (distinct from QA clustering model to eliminate circularity).

Guards against asymmetric meaning shifts not detected by embedding similarity.

G3: Negation preservation (no polarity change):

NegEx for medical instances; dependency parse + regex for general instances.

Guards against accidental negation scope introduction or removal.

G4: Grammaticality (GPT-2-medium perplexity ratio ≤ 2.0):

GPT-2-medium (345 MB; CPU inference). Ratio = perplexity(perturbation) / perplexity(original).

Guards against ungrammatical outputs from seq2seq perturbation models.

G5: Lexical divergence (≥ 1 token changed; $\leq 60\%$ of original length):

NLTK tokenisation + difflib token-level Levenshtein edit distance.

Guards against trivial identical copies or excessive rewrites.

G6: Entity preservation (medical only; target entity or synonym present):

scispaCy NER + UMLS Metathesaurus synonymy lookup.

Guards against loss of the target medical concept through over-generalisation.

Semantic Entropy Computation

Let x_0 denote the original input and $P(x_0) = \{x_1, \dots, x_8\}$ the set of eight accepted perturbations. Each of the nine inputs is processed by a model, producing an output mapped to semantic interpretation s_i . Let $S = \{s_1, \dots, s_m\}$ denote the m distinct semantic interpretations.

CUI mapping protocol (medical concept normalisation task):

Rule 1 (primary): scispaCy top-ranked CUI when linker confidence ≥ 0.70 .

Rule 2 (disambiguation): SapBERT cosine similarity when confidence < 0.70 .

Rule 3 (multi-CUI): Most specific CUI (lowest level in UMLS semantic type hierarchy).

Rule 4 (unresolvable): Outputs with no CUI at confidence ≥ 0.50 after Rule 2 classified as 'unresolved'.

Instances with more than three unresolved outputs excluded.

Rule 5 (synonym surface forms): Surface synonyms of UMLS preferred terms assigned the CUI of the matched preferred term.

QA task semantic clustering (SQuAD 2.0 and BioASQ):

Stage 1: Character-identical answers placed in the same cluster after normalisation.

Stage 2: DeBERTa-v3-large-mnli-fever-anli (distinct from G2 model) assigns bidirectional entailment ≥ 0.80 for clustering.

Entropy formula:

Empirical probability: $p(s_i) = \text{count}(s_i) / 9$

Semantic entropy: $H(x_0) = -\sum_i p(s_i) * \log_2 p(s_i)$

Normalised entropy: $H\text{-norm}(x_0) = H(x_0) / \log_2(k+1)$, where k is the number of accepted perturbations.

$H\text{-norm}(x_0)$ in $[0, 1]$.

Bootstrap 95% confidence intervals accompany all reported mean entropy values (1,000 iterations, percentile method).

Statistical Analysis Plan

All confirmatory analyses are pre-registered on OSF before any data collection or model inference.

Three confirmatory hypothesis tests correspond to the three research questions.

RQ1: Characterisation of Semantic Instability (H1):

Primary analysis: Mixed-effects regression model.

Model: $H\text{-norm}(x_0) \sim \text{PerturbationType} * \text{LinguisticCategory} + \text{Magnitude} + \text{Accuracy} + (1|\text{Instance}) + (1|\text{Model})$

Estimation: Restricted Maximum Likelihood (REML) via `statsmodels.formula.api.mixedlm` in Python or `lme4` in R.

Pre-specified minimum effect size: $f\text{-squared} \geq 0.04$ for fixed effects of interest; significant interaction term at BH-FDR $q = 0.05$.

Marginal and conditional R-squared reported following Nakagawa and Schielzeth (2013).

Fixed-effect 95% confidence intervals via parametric bootstrap (1,000 iterations).

RQ2: Accuracy-Stability Dissociation (H2):

Primary analysis: One-tailed Wilcoxon signed-rank test.

Compares $H\text{-norm}(x_0)$ distributions: top-quartile (high-accuracy) vs. bottom-quartile (low-accuracy) instances within each model.

Pre-specified threshold: rank-biserial $r \geq 0.30$ (medium effect size, Cohen 1988).

Analysed separately for encoder-only (CUI-ranking entropy) and generative (text-generation entropy) architectures.

Cross-task replication on BioASQ and SQuAD 2.0 as robustness checks (exploratory).

Global dissociation also tested via Spearman rank correlation between per-instance accuracy and $H\text{-norm}(x_0)$, pooled via Fisher's z-transformation. Pre-specified threshold: $|\rho| < 0.30$.

RQ3: Domain Adaptation and Semantic Stability (H3):

Primary analysis: One-tailed Mann-Whitney U tests in two pre-specified within-scale pairs.

Pair 1: BioBERT vs. BERT-base (both 110M parameters).

Pair 2: BioMistral-7B vs. FLAN-T5-XXL (large-scale tier).

Pre-specified threshold: rank-biserial $r \geq 0.30$ in each pair.

Generalisation condition (pre-registered): Attenuation of the domain-adaptation stability advantage on SQuAD 2.0 relative to MedMentions, tested by repeating within-scale comparisons on SQuAD 2.0 and reporting the change in effect size.

All confirmatory tests use Benjamini-Hochberg (BH) false discovery rate correction at $q = 0.05$. Both uncorrected and BH-adjusted p-values reported.

Pre-Registered Sensitivity Analyses

Six sensitivity analyses assess robustness of the main findings:

1. SA1 (perturbation count): Repeat main entropy analysis at $k = 5$ and $k = 7$ using random subsets of the eight accepted perturbations; verify H1-H3 conclusions are stable.
2. SA2 (gate threshold variation): Repeat at all six thresholds varied by ± 0.05 ; report stability of H1-H3 across the three threshold regimes.
3. SA3 (NLI-borderline exclusion): Exclude perturbations with G2 score in range 0.80-0.85 and rerun main analysis.
4. SA4 (temperature ablation): Report BioMistral-7B entropy at $T = 0.5$ and $T = 1.0$ on $n = 60$ instances to confirm greedy decoding is appropriate.
5. SA5 (CUI rule sensitivity): Repeat main analysis using Rule 1 assignments only (confidence ≥ 0.70 , no SapBERT fallback) and compare results.
6. SA6 (exclusion impact): Verify excluded instances do not systematically differ from included instances by comparing semantic type distribution, mention length, and baseline accuracy distributions.

Methodology Based on Research Questions

RQ1: Which linguistic properties of meaning-preserving perturbations are the strongest independent predictors of semantic entropy in clinical LLM outputs, and do these predictors interact?

Summary Objective

Characterise what drives semantic entropy in clinical LLM interpretation at the level of specific linguistic factors perturbation type, magnitude of lexical change, and grammatical category of modified tokens using UMLS-grounded CUI clustering as the formal equivalence criterion.

Approach

Generate $k = 8$ meaning-preserving perturbations per instance across four methods and annotate each perturbation along three linguistic dimensions:

Perturbation type: categorical label from the four methods.

Magnitude: normalised Levenshtein edit distance (continuous and binned: low < 0.20 , medium $0.20-0.40$, high > 0.40).

Linguistic category: Noun, Verb, Adjective/Adverb, Function Word (SpaCy POS tagging on original input; head token category for multi-token edits).

Process all nine inputs per instance through each of the five models.

Map outputs to UMLS CUIs using the five-rule protocol and compute $H\text{-norm}(x_0)$ per instance per model.

Fit the mixed-effects regression model: $H\text{-norm}(x_0) \sim \text{PerturbationType} * \text{LinguisticCategory} + \text{Magnitude} + \text{Accuracy} + (1|\text{Instance}) + (1|\text{Model})$.

Evaluation Metrics

Fixed-effect sizes (f-squared) for perturbation type, linguistic category, magnitude, and their interaction.

Statistical significance at BH-FDR $q = 0.05$.

Marginal R-squared (variance explained by fixed effects alone).

Conditional R-squared (variance explained by fixed and random effects combined).

Negative Control

For a subsample of 60 instances, two control inputs are generated: (1) an identical copy of the original ($H\text{-norm}$ must = 0 exactly) and (2) a typographic-only variant with capitalisation changes only ($H\text{-norm}$ must be approximately 0). Any model producing $H\text{-norm} > 0.10$ on the identical-copy control has a determinism failure and is excluded from the main analysis with documentation.

RQ2: Does semantic entropy detect systematic interpretation instability in high-accuracy LLM outputs that traditional evaluation metrics fail to reveal, and does this dissociation hold across encoder-only and generative architectures?

Summary Objective

Test whether models that perform well on task accuracy can still exhibit elevated semantic entropy specifically in the high-accuracy quartile of the output distribution demonstrating that accuracy-based evaluation systematically underestimates reliability risk in clinical NLP.

Approach

For each model, compute per-instance task accuracy (CUI match or answer span overlap) and H-norm(x0) across the full dataset.

Identify the top accuracy quartile and bottom accuracy quartile for each model.

Apply a one-tailed Wilcoxon signed-rank test comparing H-norm(x0) distributions in these two groups.

Conduct the analysis separately for encoder-only models (CUI-ranking entropy) and generative models (text-generation entropy) to test the cross-architecture scope condition.

Replicate on BioASQ and SQuAD 2.0 as exploratory cross-task robustness checks.

Evaluation Metrics

Spearman rank correlation between accuracy and H-norm(x0) per model (global dissociation, H1, pre-specified threshold $|\rho| < 0.30$).

Rank-biserial r from Wilcoxon test (pre-specified threshold $r \geq 0.30$ for H2).

95% bootstrap confidence intervals on all effect sizes.

Limitations

Accuracy-stability dissociation in the high-accuracy tail can only be interpreted as clinically meaningful if the effect size meets the pre-specified threshold statistical significance alone is not sufficient given $n = 550$.

Cross-architecture comparisons are presented as supplementary analysis only, with the architectural confound annotated.

RQ3: Does domain-specific biomedical pretraining reduce semantic entropy relative to general-purpose models at equivalent parameter scale, and does this advantage generalise across task formats?

Summary Objective

Test whether domain adaptation (biomedical pretraining) confers stability advantages beyond accuracy improvements, using a within-scale comparison design that isolates pretraining from model scale. The pre-registered generalisation condition tests whether any advantage is task-specific or fully general.

Approach

Within-scale comparison 1: BioBERT vs. BERT-base (both 110M parameters).

Within-scale comparison 2: BioMistral-7B vs. FLAN-T5-XXL (large-scale tier).

Primary evaluation on MedMentions ST21pv (concept normalisation).

Generalisation evaluation on BioASQ (biomedical QA) and SQuAD 2.0 (general-domain QA).

Cross-scale comparisons (e.g., BioMistral-7B vs. BioBERT) presented as supplementary analysis only, with the scale confound explicitly annotated.

Evaluation Metrics

Rank-biserial r from Mann-Whitney U test (one-tailed; pre-specified threshold $r \geq 0.30$ in each within-scale pair).

Change in effect size from MedMentions to SQuAD 2.0 for the generalisation condition.

95% bootstrap confidence intervals on all reported entropy means and effect sizes.

Ethical Considerations

Patient Data Privacy and Dataset Licensing

All three datasets (MedMentions ST21pv, SQuAD 2.0, BioASQ) are publicly available de-identified research datasets containing no patient identifiable information. MedMentions uses PubMed abstracts which are public domain. SQuAD 2.0 is released under CC BY-SA 4.0. BioASQ is available to registered participants under the challenge data agreement. No patient data or private clinical records are accessed at any stage of the research. Confidentiality is always maintained through the exclusive use of these de-identified, publicly licensed datasets. Informed consent from individual participants is not required because no human participants are recruited, surveyed, or involved in any experimental procedure the study uses only pre-existing publicly released datasets with appropriate consent and

licensing mechanisms established by their original creators. No individual informed consent process is therefore required.

Clinical Safety and Hallucination Risk

The outputs of LLMs considered in this study will not be used in the context of any clinical decision, diagnosis, or patient care in any situation but will be the basis of research measurement. The paper is a direct response to the clinical safety issue that high accuracy standards may be coupled with semantically unstable results identified in P. Hager et al. (2024, Nature Medicine) and M. Agrawal et al. (2025, npj Digital Medicine). The entropy metric is confirmed in the negative control condition (identical-copy perturbation) prior to any model comparisons taking place.

Reproducibility and pre-registration

Every confirmatory (H1, H2, H3) and sensitivity analysis (SA1-SA6) will be pre-registered in OSF, prior to data collection or model inference. The pre-registration identifier will be listed in the final thesis. All code is released on GitHub under MIT licence with fixed random seeds (numpy: 42; PyTorch: 42; HuggingFace Transformers: 42) and pinned model revision hashes. Bit-identical reproduction is possible with a Docker container image that is pinned to specific dependencies. Entire additional data perturbation datasets, gate scores, CUI mappings, model outputs, and entropy scores are published publicly.

Scope Limitations and English-Centric Focus

This research paper is limited to English-language clinical text. This is recognized as a scope condition in the research questions. As A. Nevéol et al. (2018) and W. Boag et al. (2025) comment, the majority of clinical NLP and consistency research is English-oriented, and the current study is silent on multilingual clinical NLP. The results might not be applicable to non-English clinical language environments.

Confidentiality and Informed Consent

The research is conducted in a confidential manner by using de-identified and publicly published data only. None of the personally identifiable data is accessed, stored or processed at any point. No informed consent is necessary since all the data utilized in this study will be sourced publicly available research datasets (MedMentions, SQuAD 2.0, BioASQ) that have adequate consent and licensing mechanisms in place by their authors. No human subjects are recruited, surveyed or subjected to any experimental procedure. The research does not entail any communication with anyone and does not produce any

personally identifiable results. No specific informed consent procedure is thus necessary and no HREC ethics approval is necessary except compliance with the dataset licensing agreements already in place.

Responsible Use of LLMs

All the inferences of the LLM will be done in a controlled research environment. Models are not tools that are deployed. There is no participant recruitment. There is no clinical advice generated. Model inference does not involve any human participants. The study does not involve human participation, except the research team that would be involved in data processing and analysis. As a result, there is no need to have HREC (Human Research Ethics Committee) approval besides dataset licensing agreements that are already in place.

Limitations of Selected Methods

Perturbation Quality and Coverage

The six-gate automated pipeline has a calibrated false-admission rate of $\leq 10\%$ on negative calibration pairs (deliberately similar concept pairs from different CUIs). This is within the 10-15% inter-annotator disagreement range typical in biomedical entity annotation, but it is not zero some non-equivalent perturbations may pass all gates.

Back-translation and paraphrase models are general-domain seq2seq models not specifically trained on clinical text. Perturbation quality for rare clinical terminology may be lower than for common medical concepts.

Syntactic reordering is constrained to transitive verbs with explicit subjects and objects, which limits its application to a subset of clinical sentences.

Model Inference Constraints

FLAN-T5-XXL (11B parameters) and BioMistral-7B (7B parameters) are loaded with INT8 quantisation, which may introduce minor numerical differences compared to full-precision inference. A temperature ablation sensitivity analysis (SA4) tests whether this affects the entropy results.

Greedy decoding ($T = 0.0$) eliminates sampling stochasticity but means all observed entropy reflects input variation rather than output randomness. This is methodologically appropriate for the study's purpose but means results are not directly comparable to studies using sampling-based entropy estimation.

The study does not fine-tune any models on the evaluation tasks. Zero-shot performance is the methodological choice this preserves the study's generalisability but means models may perform below their task-specific ceiling.

Statistical Analysis Constraints

The mixed-effects regression for RQ1 assumes approximate normality of residuals. If residuals are substantially non-normal, a non-parametric permutation-based alternative will be applied and documented as a sensitivity deviation from the pre-registered protocol.

The within-scale comparison for RQ3 is limited to two pairs. Cross-scale and cross-architecture comparisons are exploratory only and must not be interpreted as confirmatory evidence.

The study targets $n = 550$ instances from MedMentions with a 10% buffer for quality-gate rejections. If rejection rates exceed this buffer, the analysis will proceed with the available n and the shortfall will be documented.

Key Milestones and Timeline

The following milestones mark the major stages of the research project. Deadlines are set to allow supervisor review at each stage before proceeding to the next.

Milestone	Deliverable	Notes and Dependencies
M1: Pre-registration	OSF pre-registration of all confirmatory analyses and sensitivity analyses. Deadline: Before any data collection.	• Must be completed before M2. Pre-registration identifier recorded in thesis. • Covers H1, H2, H3, SA1-SA6, sample size justification, and CUI mapping protocol.
M2: Dataset preparation	Stratified sampling of MedMentions ($n=550$), SQuAD 2.0 ($n=200$), BioASQ ($n=150$). Pilot set reserved ($n=55+20+15$).	• scispaCy UMLS linker validation on full MedMentions for inclusion criteria. • Pilot instances excluded from main analysis.

Milestone	Deliverable	Notes and Dependencies
	Deadline: Week 2 of data collection phase.	Stratification verified across semantic type categories.
M3: Threshold calibration	Six-gate thresholds calibrated on 200 positive and 200 negative UMLS proxy pairs. Deadline: Before full perturbation generation.	- Calibration uses pilot set only. Results documented in supplementary materials. - SA2 threshold variation analysis depends on this calibration data.
M4: Perturbation generation	k=8 perturbations generated and validated for all instances across three datasets. Deadline: 2 weeks after M3.	- Rejection rate logged per gate per method. - Up to 3 regeneration attempts per slot. - Estimated storage: ~15 GB for perturbation datasets.
M5: Negative control validation	Identical-copy and typographic-only control results verified for all five models. Deadline: Before main inference.	- Any model with H-norm > 0.10 on identical-copy control excluded with documentation. - Results presented as the first finding in Results chapter.
M6: Model inference	All five models process all nine inputs per instance across all three datasets. Deadline: 2 weeks after M5.	- Encoder-only on GPU-1; generative on GPU-2 (INT8). - Estimated wall time: 26-43 hours total. - CUI mappings and output logs saved to SSD.
M7: Entropy computation	H-norm(x0) computed for all instances and models.	Bootstrap CIs computed for all reported mean entropy values.

Milestone	Deliverable	Notes and Dependencies
and cross-linker check	MetaMap cross-linker reliability check ($\kappa \geq 0.75$ target) on 10% subsample. Deadline: 1 week after M6.	Cross-linker results reported in Results as pipeline reliability metric.
M8: Confirmatory statistical analyses	H1, H2, H3 tests conducted. BH-FDR correction applied. Sensitivity analyses SA1-SA6 conducted. Deadline: 2 weeks after M7.	- Any deviations from pre-registered protocol documented as exploratory. - SA6 exclusion impact analysis requires comparison of included vs. excluded instance characteristics.
M9: Draft results chapter	Complete results chapter submitted to supervisor team for review. Deadline: 1 week after M8.	- Tables and figures for all H1-H3 tests. - Sensitivity analysis results. - Negative control and cross-linker reliability results.
M10: Code and data release	GitHub repository with pinned checkpoints, fixed seeds, and Docker image. Public data repository with all perturbation datasets and entropy scores. Deadline: Concurrent with thesis submission.	- MIT licence for code. - Docker image ensures bit-identical reproduction of results. - Pre-registration identifier linked in README.

Hardware and Infrastructure Constraints

Two NVIDIA RTX A4000 GPUs (16 GB VRAM each) are available. FLAN-T5-XXL (11B INT8) and BioMistral-7B (7B INT8) each require approximately 11 GB and 8 GB VRAM respectively, fitting within GPU-2 capacity sequentially but not simultaneously.

Wall time is estimated at 26-43 hours across all inference tasks. GPU availability on shared infrastructure could extend this timeline; buffer time is included in the milestone schedule.

Storage: approximately 49 GB of the 200 GB SSD will be used for model checkpoints, perturbation datasets, output logs, and entropy scores.

References:

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